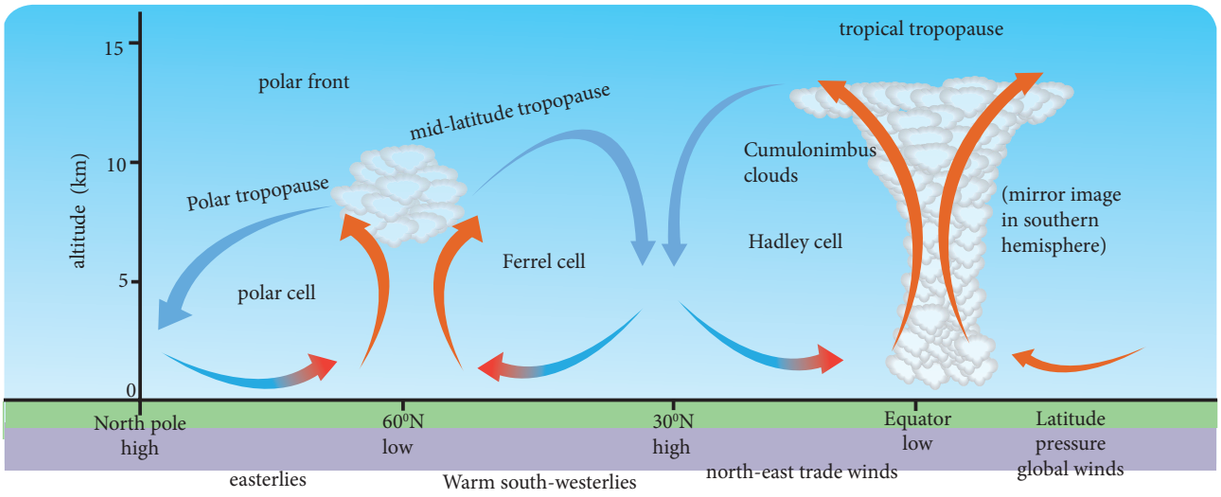


Figure 6.14 Meridional circulation

Shifting of Pressure Belts and Primary Wind System

These pressure belts and primary wind systems are dynamic in character as they shift 5° north and 5° south from their position along with the apparent movement of the sun.

6.4.8 Secondary Wind System

Both monsoon and cyclones are considered as secondary or regional wind systems.

Monsoons

The word 'Monsoon' is derived from the Arabic word, 'Mausim' which means 'Season'. Monsoons are seasonal winds which reverse their direction due to unequal heating and cooling of the land and the water.

Mechanism of Monsoon

The land absorbs more heat energy during summer, which leads to the formation of low pressure over continent. But the ocean will have relatively lower temperature than the continent leading to the formation of high pressure system over ocean. So, the wind blows from sea to land during summer season. Meanwhile the land reradiates more heat energy to space during winter leading to the formation of high pressure above the continent. But the ocean will have relatively higher temperature than the continent leading to formation of low pressure system over ocean. So, wind blows from land to sea during the winter season. This mechanism has an important effect on rainfall received over the region.

Nature of Monsoon System

There are three distinct characteristics related to monsoon wind system which differentiates it from other wind systems. They are;

1. Minimum 160° reversal of wind direction between seasons.
2. They affect a large part of the continents and oceans.
3. The formation of low and high pressure systems over land and water and their interchange between the seasons.

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